

Dr. Doc:

Protein Structure and Organization

Levels of Protein Organization

Proteins are organized into four structural levels: primary, secondary, tertiary, and quaternary structures. Each level builds on the previous one and determines the protein's shape, stability, and biological function.

▼ **Primary Structure:** The primary structure is the linear sequence of amino acids linked by peptide bonds, extending from the N-terminus to the C-terminus. It forms the foundation of all higher protein structures, and even a single amino acid change can affect protein function.

▼ **Secondary Structure:** The secondary structure is the local folding of the polypeptide chain into α -helices, β -pleated sheets, and random coils, stabilized mainly by hydrogen bonds.

▼ **Tertiary Structure:** The tertiary structure is the complete three-dimensional folding of a single polypeptide chain. This folding produces the functional form of many proteins and is stabilized by several types of bonding interactions.

▼ **Quaternary Structure:** The quaternary structure is formed when two or more separate polypeptide chains (subunits) assemble to form one functional protein. This level is present only in proteins composed of multiple polypeptide chains.

Tertiary Structure

Characteristics of Functional Globular Proteins

- ▼ Functional globular proteins are compact, spherical, and generally water-soluble, allowing efficient function in body fluids.
- ▼ Hydrophobic amino acid residues are buried inside the protein, while hydrophilic residues remain on the surface and interact with water.
- ▼ The unique three-dimensional shape forms enzyme active sites, receptor sites, and binding sites essential for protein function.
- ▼ Disruption of the tertiary structure by denaturation can cause the protein to lose its normal function.

Bonding Interactions that Stabilize Tertiary Structure

- ▼ Hydrogen bonds: Stabilize folded regions through weak attractions between polar groups.
- ▼ Ionic bonds: Form between charged amino acid side chains and help maintain protein shape.
- ▼ Hydrophobic interactions: Keep nonpolar residues inside the protein away from water.
- ▼ Van der Waals forces: Provide weak attractions between closely packed atoms.
- ▼ Disulfide bonds: Strong covalent bonds between cysteine residues that increase protein stability.

Quaternary Structure

Assembly of Separate Polypeptide Chains (Subunits)

▼ Quaternary structure is produced by the association of two or more independently folded polypeptide chains, called subunits, into a single functional protein.

▼ The subunits are held together by hydrogen bonds, ionic bonds, hydrophobic interactions, Van der Waals forces, and sometimes disulfide bonds, rather than by peptide bonds.

▼ The interaction among subunits often increases the protein's stability and improves its biological function.

Oligomeric (Multimeric) Proteins

▼ Proteins composed of multiple subunits are called oligomeric or multimeric proteins.

▼ According to the number of subunits, they are classified as dimers (2 subunits), trimers (3 subunits), tetramers (4 subunits), pentamers (5 subunits), or multimers (many subunits).

▼ A common example is hemoglobin, which is a tetramer composed of four polypeptide chains that work together to transport oxygen.

Homogeneous vs. Heterogeneous Quaternary Structure

▼ Homogeneous (Homomeric) proteins contain identical subunits, meaning all polypeptide chains are the same.

▼ Heterogeneous (Heteromeric) proteins contain different types of subunits, each contributing a specific role to the protein's function.

▼ Hemoglobin ($\alpha_2\beta_2$) is a classic example of a heteromeric protein, consisting of two alpha (α) and two beta (β) polypeptide chains.

💡 Key Point: The primary structure determines how a protein folds into secondary, tertiary, and quaternary structures. The final three-dimensional structure ultimately determines the protein's stability, activity, and biological function.

Myoglobin and Hemoglobin Overview

🔥 Biological Roles

Myoglobin and hemoglobin are oxygen-binding proteins, but they perform different functions in the body. Myoglobin acts as an oxygen storage protein in skeletal muscle, supplying oxygen during periods of increased muscular activity. In contrast, hemoglobin functions as an oxygen transport protein, carrying oxygen from the lungs to body tissues and returning carbon dioxide from tissues to the lungs.

🎯 Locations

▼ Myoglobin is found in the cytoplasm of skeletal (striated) muscle cells, where it stores oxygen for muscle metabolism.

▼ Hemoglobin is found only inside erythrocytes (red blood cells), where it transports oxygen throughout the body.

🌟 Structural Composition

▼ Myoglobin is a monomeric protein, consisting of one polypeptide chain with a single heme group, allowing it to bind one oxygen molecule.

▼ Hemoglobin is a tetrameric metalloprotein, composed of four polypeptide chains (2 α and 2 β in adults), each containing a heme group with an iron (Fe^{2+}) atom. This enables one hemoglobin molecule to bind up to four oxygen molecules simultaneously.

✦ Key Point: Myoglobin stores oxygen in muscles, whereas hemoglobin transports oxygen in the blood. Myoglobin is a monomer, while hemoglobin is a tetrameric metalloprotein.

🔥 Globular Hemeproteins and the Heme Group

✦ Definition of Hemeproteins

▼ Hemeproteins are a group of specialized globular proteins that contain a tightly bound heme prosthetic group. The heme group gives these proteins their ability to bind oxygen and carry out important biological functions such as oxygen storage, oxygen transport, and electron transfer reactions.

✦ Structure of Heme

▼ The heme group is an iron-containing molecule composed of ferrous iron (Fe^{2+}) surrounded by Protoporphyrin IX.

▼ Protoporphyrin IX consists of four pyrrole rings connected by methene bridges and contains methyl groups, vinyl groups, and propionate side chains that contribute to the structure and function of heme.

✦ Coordination Sites of Heme Iron

▼ The ferrous iron (Fe^{2+}) in heme has six coordination sites that allow interaction with the surrounding molecules.

▼ 5th coordination site: The iron is attached to the imidazole ring of proximal histidine, which connects the heme group to the protein structure.

▼ 6th coordination site: This position is available for oxygen binding, allowing hemeproteins to carry or store oxygen.

✦ Role of Histidine

▼ Proximal histidine: Binds directly to the iron atom and stabilizes the position of the heme group within the protein.

▼ Distal histidine: Helps stabilize oxygen binding, improves oxygen affinity, and protects the iron atom from oxidation, ensuring proper function of the hemeprotein.

🔥 Detailed Structure of Myoglobin

✦ Helical Regions

▼ Myoglobin is a compact globular protein in which about 80% of the polypeptide chain is folded into eight α -helices (A–H). These helices form a stable structure around the heme group, allowing myoglobin to bind and store oxygen efficiently.

✦ Amino Acid Distribution

▼ Interior: The core of myoglobin contains mainly nonpolar, hydrophobic amino acids such as valine, leucine, and phenylalanine. These amino acids are closely packed together and help maintain the stability of the protein structure.

▼ Surface: The outer surface contains charged and hydrophilic amino acids that interact with water through hydrogen bonds, making myoglobin soluble in muscle cells.

✦ Microenvironment

▼ The globin portion of myoglobin creates a specific environment around the heme group, allowing reversible oxygen binding. This environment protects the iron atom and helps myoglobin perform its role as an oxygen-storage protein in skeletal muscle.

★ Key Point: The unique α -helical structure and amino acid arrangement of myoglobin provide stability and enable efficient oxygen storage.

🔥 Detailed Structure of Hemoglobin (Hb)

💡 Hemoglobin A Subunits

▼ Hemoglobin A (HbA) is the major form of adult hemoglobin and is composed of four polypeptide chains: two α (alpha) chains and two β (beta) chains ($\alpha_2\beta_2$). Each subunit contains a heme group that allows hemoglobin to bind oxygen.

💡 Quaternary Arrangement

▼ Hemoglobin has a tetrameric quaternary structure formed by the association of two similar dimers: $(\alpha\beta)_1$ and $(\alpha\beta)_2$. These dimers are arranged together to create the functional hemoglobin molecule.

💡 Subunit Interactions

▼ The subunits of hemoglobin are held together by non-covalent interactions, including hydrophobic interactions, ionic bonds, and hydrogen bonds. These interactions provide stability and allow the subunits to interact during oxygen binding and release.

💡 Hemoglobin Concentrations

▼ Normal hemoglobin concentration varies between individuals. In general, normal levels are approximately 14–16 g/dL in males and 13–15 g/dL in females.

★ Key Point: Hemoglobin's four-subunit structure allows it to efficiently transport oxygen from the lungs to tissues and regulate oxygen delivery according to the body's needs.

🔥 Oxygen Binding and Dissociation Curves

💡 Oxygen Dissociation Curve Definition

▼ The oxygen dissociation curve is a graph that shows the relationship between oxygen saturation (Y-axis) and partial pressure of oxygen (pO_2) (X-axis). It describes how proteins like myoglobin and hemoglobin bind and release oxygen at different oxygen pressures.

💡 Myoglobin vs. Hemoglobin Curves

▼ Myoglobin: Shows a hyperbolic curve because it has a high oxygen affinity and can bind oxygen even at low pO_2 . This makes myoglobin effective for oxygen storage in muscle cells.

▼ Hemoglobin: Shows a sigmoidal (S-shaped) curve due to cooperative binding, where binding of one oxygen molecule increases the affinity for additional oxygen molecules. This allows efficient oxygen uptake in the lungs and release in tissues.

💡 P50 Values

▼ P50 is the oxygen pressure at which a protein is 50% saturated with oxygen and indicates oxygen affinity.

▼ Myoglobin: Has a low P50 value of about 1 mm Hg, showing high oxygen affinity.

▼ Hemoglobin: Has a P50 value of about 26 mm Hg, showing lower oxygen affinity and allowing oxygen release to tissues.

🌟 Key Point: Myoglobin is designed for oxygen storage, while hemoglobin is designed for oxygen transport and delivery.

🔥 Cooperative Binding and Conformational States

🌟 Cooperative Binding

▼ Cooperative binding is the process in which the binding of one oxygen molecule to hemoglobin increases the affinity of the remaining subunits for oxygen. This phenomenon, known as heme-heme interaction, allows hemoglobin to bind oxygen more efficiently.

🌟 Structural States of Hemoglobin

▼ T (Taut or Deoxygenated) State: The T state has low affinity for oxygen and is the form that readily releases oxygen to body tissues.

▼ R (Relaxed or Oxygenated) State: The R state has high affinity for oxygen and is formed after oxygen binds, allowing hemoglobin to bind additional oxygen molecules more easily.

🌟 Oxygenation Mechanism

▼ When oxygen binds to the heme iron (Fe^{2+}), the iron atom is pulled into the plane of the heme ring. This movement causes conformational changes in the globin protein, converting hemoglobin from the T state to the R state and increasing its oxygen-binding affinity.

★ Key Point: Cooperative binding enables hemoglobin to load oxygen efficiently in the lungs and release it effectively in the tissues through the transition between the T and R states.

🔥 Allosteric Effects and Regulators

💡 Hemoglobin as an Allosteric Protein

▼ Hemoglobin is an allosteric protein, meaning its oxygen-binding affinity is regulated by molecules that bind to sites other than the oxygen-binding (heme) site. These allosteric modulators alter hemoglobin's structure and influence oxygen loading and unloading.

💡 2,3-Bisphosphoglycerate (2,3-BPG)

▼ 2,3-BPG is an intermediate produced during glycolysis in red blood cells (RBCs).

▼ It binds to the central pocket of deoxyhemoglobin (T-state), stabilizing the T state and reducing hemoglobin's affinity for oxygen, thereby promoting oxygen unloading to tissues.

▼ Clinical significance: Levels of 2,3-BPG increase in conditions such as high altitude, chronic pulmonary disease, and severe anemia, helping improve oxygen delivery to tissues. It also helps maintain oxygen-release capacity in stored blood.

💡 The Bohr Effect (pH and CO₂)

▼ The Bohr effect states that increased H⁺ concentration (low pH) and high pCO₂ decrease hemoglobin's affinity for oxygen, causing the oxygen dissociation curve to shift to the right and promoting oxygen release to tissues.

▼ Carbon dioxide (CO_2) is transported in the blood in three forms: dissolved in plasma, bound to hemoglobin as carbaminohemoglobin, and mainly as bicarbonate (HCO_3^-) ions.

✦ Temperature

▼ Increased temperature decreases hemoglobin's affinity for oxygen, causing a rightward shift of the oxygen dissociation curve and enhancing oxygen unloading to active tissues, especially during exercise.

★ Key Point: 2,3-BPG, increased H^+ , increased CO_2 , and high temperature all decrease hemoglobin's oxygen affinity, promoting oxygen release where it is most needed.

🔥 Hemoglobin Derivatives and Clinical Significance

✦ Carboxyhemoglobin (COHb)

▼ Carboxyhemoglobin (COHb) is formed when carbon monoxide (CO) binds to hemoglobin. CO has about 200 times greater affinity for hemoglobin than oxygen, preventing oxygen transport and causing carbon monoxide poisoning.

✦ Methemoglobin (MetHb)

▼ Methemoglobin (MetHb) is formed when the iron in heme is oxidized from ferrous (Fe^{2+}) to ferric (Fe^{3+}) state. Since Fe^{3+} cannot bind oxygen, methemoglobin is unable to transport oxygen effectively, leading to reduced oxygen delivery to tissues.

✦ Glycosylated Hemoglobin (HbA1c)

▼ HbA1c is formed when glucose binds irreversibly to hemoglobin. It reflects the average blood glucose level over the previous 2–3 months and is widely used to monitor and assess long-term glucose control in diabetes mellitus.

💡 Fetal Hemoglobin (HbF)

▼ Fetal hemoglobin (HbF) has the structure $\alpha_2\gamma_2$, containing two alpha and two gamma chains.

▼ HbF has reduced affinity for 2,3-BPG, giving it a higher affinity for oxygen than adult hemoglobin. This allows efficient transfer of oxygen from the mother's blood to the fetus across the placenta.

🌟 Key Point

▼ COHb reduces oxygen transport due to CO poisoning, MetHb cannot bind oxygen because of Fe^{3+} , HbA1c is an indicator of long-term blood glucose control, and HbF has high oxygen affinity to facilitate oxygen transfer from mother to fetus.

🔥 Hemoglobinopathies and Thalassemia

💡 Definition

▼ Hemoglobinopathies are genetic disorders caused by mutations that result in abnormal hemoglobin structure or reduced hemoglobin synthesis, leading to impaired oxygen transport and anemia.

💡 Sickle Cell Anemia (HbS)

▼ Sickle cell anemia (HbS) is caused by the replacement of glutamic acid with valine at the 6th position of the β -globin chain.

▼ This mutation causes red blood cells to become sickle-shaped, leading to hemolysis, anemia, and blockage of small blood vessels.

✦ Hemoglobin C Disease (HbC)

▼ Hemoglobin C disease (HbC) results from the replacement of glutamic acid by lysine in the β -globin chain.

▼ The abnormal hemoglobin forms hexagonal crystals inside red blood cells, causing mild hemolytic anemia.

✦ Thalassemias

▼ Thalassemias are inherited disorders characterized by decreased synthesis of α - or β -globin chains.

▼ The imbalance in globin chain production causes hemolysis, chronic anemia, and reduced oxygen-carrying capacity.

✦ Key Point

▼ Hemoglobinopathies involve abnormal hemoglobin structure (e.g., HbS and HbC), whereas thalassemias result from decreased synthesis of normal globin chains, leading to hemolysis and anemia.

🔥 Lung Surfactant

✦ Composition

▼ Lung surfactant is a complex mixture composed of 90% phospholipids and 10% surfactant proteins. Its major phospholipid is Dipalmitoylphosphatidylcholine (DPPC/Lecithin), while the surfactant proteins include SP-A, SP-B, SP-C, and SP-D, which work together to maintain normal lung function.

✦ Synthesis

▼ Lung surfactant is synthesized and secreted by Alveolar Type II cells in the alveoli, where it is continuously released to maintain proper lung expansion.

✦ Biological Roles

▼ Lung surfactant reduces alveolar surface tension, preventing alveolar collapse (atelectasis) during expiration. It also provides innate immune defense, as SP-A and SP-D act as collectins that help eliminate microorganisms. In addition, surfactant improves airway clearance and keeps the airways dry, promoting efficient gas exchange.

✦ Clinical Significance

▼ Infant Respiratory Distress Syndrome (IRDS) occurs mainly in preterm infants due to surfactant deficiency, resulting in breathing difficulty. The Lecithin–Sphingomyelin (L/S) ratio is used to assess fetal lung maturity, with a high ratio indicating mature lungs and a low ratio indicating a risk of IRDS.

▼ Acute Respiratory Distress Syndrome (ARDS) occurs when surfactant is damaged or inactivated, leading to impaired gas exchange and respiratory failure. Surfactant abnormalities may result from lung immaturity, lung injury, or destruction by proteases and phospholipases, reducing its normal function.

★ Key Point

▼ Lung surfactant, produced by Alveolar Type II cells, is essential for reducing surface tension, preventing atelectasis, supporting innate immunity, and maintaining efficient respiration.

🔥 Pulmonary Anatomy and Gas Exchange

💡 Alveolar Structure

▼ Alveoli are tiny air sacs in the lungs that serve as the primary site of gas exchange. Each alveolus is surrounded by a dense network of capillaries, forming the blood-air (alveolar-capillary) barrier, which is thin to allow rapid diffusion of gases.

💡 Mechanism of Gas Exchange

▼ Gas exchange occurs by simple diffusion across the blood-air barrier. Oxygen (O_2) diffuses from the alveoli into the pulmonary capillaries, where it binds to hemoglobin, while carbon dioxide (CO_2) diffuses from the blood into the alveoli to be exhaled.

🌟 Key Point

▼ The thin blood-air barrier and rich capillary network enable efficient oxygen uptake and carbon dioxide removal, ensuring effective respiration.